

Variable	Type of variable	Model used to predict missing data*	Missing value %
Mother's place of birth	Binary	Logistic regression	0.1%
Health insurance coverage	Binary	Logistic regression	0.5%
Smoking before pregnancy	Binary	Logistic regression	0.5%
Smoking during pregnancy	Binary	Logistic regression	0.8%
Living with a partner	Binary	Logistic regression	0.6%
Parity	Binary	Logistic regression	0.3%
Number of ultrasounds	Binary	Logistic regression	3%
Restrictive diet before pregnancy	Binary	Logistic regression	12.4%
Restrictive diet during pregnancy	Binary	Logistic regression	15.3%
Gestational diabetes	Binary	Logistic regression	4.3%
Hypertension	Binary	Logistic regression	2.1%
Psychological difficulties during pregnancy	Binary	Logistic regression	0.6%
Hospitalization during pregnancy	Binary	Logistic regression	0.8%
Premature rupture of membranes	Binary	Logistic regression	2.7%
Maternal medical history that can influence weight variations	Binary	Logistic regression	6.1%
Maternal medical history that could influence fetal growth	Binary	Logistic regression	6.1%
Women with child ≤ 2 years	Binary (yes/no)	Logistic regression	0%
Child's sex	Binary	Logistic regression	0.1%
Maternal age	Continuous	Linear regression	0.02%
Maternal weight	Continuous	Linear regression	1.3%
Maternal height	Continuous	Linear regression	0.03%
Body mass index	Continuous	Linear regression	1.3%
Gestational weight gain	Continuous	Linear regression	2%
Gestational age at birth	Continuous	Linear regression	0.8%
Birth weight	Continuous	Linear regression	1.4%
Birth length	Continuous	Linear regression	9%
Head circumference at birth	Continuous	Linear regression	9.7%
Ultrasonography data second trimester (Z-scores)	Continuous	Linear regression	25-69%**
Ultrasonography data third trimester (Z-scores)	Continuous	Linear regression	20-69%**
Weight variations before pregnancy	Categorical (3 categories)	Multinomial regression	13.2%
Activity status	Categorical (3 categories)	Multinomial regression	2.4%
Maternal level of education	Categorical (4 categories)	Multinomial regression	0.01%
Marital status	Categorical (3 categories)	Multinomial regression	2%
Socio-professional category	Categorical (5 categories)	Multinomial regression	0%
Complementary health assurance	Categorical (3 categories)	Multinomial regression	2.1%
Medical history of diabetes	Categorical (4 categories)	Multinomial regression	2.1%
Medical history of hypertension	Categorical (3 categories)	Multinomial regression	1.4%
Pregnancy caregiver	Categorical (4 categories)	Multinomial regression	0.9%
Prenatal consultations	Ordinal (3 categories)	Multinomial regression	1.8%
Consumption of alcohol	Categorical (4 categories)	Multinomial regression	0.7%
Maternity size strata	Categorical (5 categories)	Multinomial regression	0%
Period of inclusion	Categorical (4 categories)	Multinomial regression	0%
Mode of birth	Categorical (3 categories)	Multinomial regression	1.9%